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Bias–Variance Decomposition and the Double Descent Phenomenon

Phillip Korolev[†,*], Connor Bos[†], Yimeng Fang[†], Lim Li Sheng Isaiah[†]
[†] University of Ottawa

Abstract

The bias-variance decomposition is a foundational result in statistical learning theory that decomposes the expected prediction error of an estimator into three components: squared bias, variance, and noise. This paper aims to derive the classical decomposition, explore its connection to the bias-variance and approximation-estimation error tradeoffs, and validate theoretical results through polynomial regression experiments. Then, we extend these topics into over-parametrized regimes, where the classical U-shaped test error curve gives way to the *double descent* phenomenon first documented by Belkin et al. (2019). Through polynomial regression and the Gaussian feature models, we reproduce the double descent trend, investigate the role of implicit and explicit regularization, and summarize the effects of model parameters on the test error's second descent.

Keywords: Bias-variance decomposition, bias-variance tradeoff, polynomial regression, double descent, over-parameterization

1. Introduction

One of the more central problems in supervised learning is choosing a model that generalizes well to unseen data. The classical answer is provided by the bias-variance decomposition, which partitions the expected prediction error of an estimator into three components: the squared bias, reflecting the systematic error from a particular model; the variance, which captures sensitivity to a given training set; and irreducible noise [2, 4]. This decomposition suggests that test error follows a U-shaped curve as model complexity increases – simple models underfit (high bias), complex models overfit (high variance), while the optimal model balances the two. However, an observation by Belkin et al. [1] challenged this classical picture. When model complexity is pushed beyond the *interpolation threshold* (the point at which the model has enough parameters to fit the training data exactly), test error does not continue to rise. Instead, it begins to decrease once again, producing what is known as the *double descent* phenomenon.

This paper presents an exploration and analysis of both frameworks. We begin by deriving the classic bias-variance decomposition and establishing its connection to the approximation-estimation error tradeoff (Sections 2 and 3). We then validate the decomposition computationally through polynomial regression experiments, illustrating the classic U-shaped curve and examining the role of experimental parameters. Finally, we extend the analysis into over-parametrized regimes, reproducing double descent in both polynomial regression and the Gaussian feature model. We investigate the mitigating role of explicit regularization, and discuss the implicit regularization process that drives the second descent (Section 5).

2. The Bias–Variance Decomposition

2.1. General Setting

We begin by deriving the classical bias-variance decomposition for some arbitrary estimator. Recall, the *bias* of our estimator essentially measures the difference between the expected value of the estimator and the true value. On the other hand, we can describe

[*] pkoro029@uottawa.ca, cbos035@uottawa.ca, yfang077@uottawa.ca, llim097@uottawa.ca

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variance as the quantity that our model's estimation varies depending on its setting and the data set which we draw. Consider an unknown quantity θ and an estimator $\hat{\theta}$ computed from data.

Definition 2.1 (Bias and Variance of an Estimator)**.** Let $\hat{\theta}$ be an estimator of a deterministic quantity θ. The *bias* of $\hat{\theta}$ is defined as $\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$, and the *variance* is $\mathrm{Var}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2]$.

Then, we can decompose the mean squared error (MSE) of the estimator as follows:

Theorem 2.2 (Bias–Variance Decomposition for MSE)**.** *For any estimator $\hat{\theta}$ of a deterministic quantity θ,*

$$\mathbb{E}\Big[(\hat{\theta} - \theta)^2\Big] = \big(\mathbb{E}[\hat{\theta}] - \theta\big)^2 + \mathrm{Var}(\hat{\theta}). \tag{2.1}$$

That is, $\mathrm{MSE}(\hat{\theta}) = \mathrm{Bias}^2(\hat{\theta}) + \mathrm{Var}(\hat{\theta})$.

Proof. We add and subtract $\mathbb{E}[\hat{\theta}]$ inside the square:

$$\begin{aligned}
\mathbb{E}\Big[(\hat{\theta} - \theta)^2\Big] &= \mathbb{E}\Big[\big((\hat{\theta} - \mathbb{E}[\hat{\theta}]) + (\mathbb{E}[\hat{\theta}] - \theta)\big)^2\Big] \\
&= \mathbb{E}\Big[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\Big] + 2\,(\mathbb{E}[\hat{\theta}] - \theta)\,\underbrace{\mathbb{E}[\hat{\theta} - \mathbb{E}[\hat{\theta}]]}_{=0} + (\mathbb{E}[\hat{\theta}] - \theta)^2 \\
&= \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}^2(\hat{\theta}).
\end{aligned}$$

□

2.2. **Classical Example Regarding Sample Mean and Variance**

To illustrate the bias-variance decomposition in a familiar statistical setting, we consider two classical estimators: the sample mean and the sample variance.

Let $X_1, ..., X_n$ be i.i.d. samples with mean μ. The sample mean is defined as

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

From $\mathbb{E}[\bar{X}] = \mu$, we know that the estimator is unbiased (Bias $= 0$) and its mean squared error is entirely made up by the variance, i.e.: $\mathrm{MSE}(\bar{X}) = \mathrm{Var}(\bar{X})$. This is a case where the entire MSE is purely driven by variance.

Now, consider the estimation of the population variance σ^2. A natural, but naive, estimator is $\sigma^2_{\text{naive}} = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2$. However, this estimator is biased. A standard correction yields the unbiased version, known as Bessel's correction:

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2,$$

which ensures $\mathbb{E}[S^2] = \sigma^2$. By the bias-variance decomposition, we then have $\mathrm{MSE}(S^2) = \mathrm{Var}(S^2)$ since it is unbiased. In this case, again, all MSE arises purely from variance across samples.

The sample mean and sample variance are included as classical benchmarks to illustrate the decomposition in a familiar setting. The sample mean, as well as the corrected sample variance, demonstrate pure variance-dominated estimators where the MSE is made up entirely of their variance. Together, these examples show that bias and variance are distinct aspects of risk relative to the Baye's optimal predictor; while bias reflects systematic model inaccuracy, variance suggests how sensitive our model is to a particular training set. Reducing one often comes at the cost of increasing the other, which is exactly the fundamental trade-off we discuss in Section 2.

2.3. Application to Regression with Squared Loss

We apply this decomposition in the context of supervised learning. Let us suppose that the data is generated by the following:

$$Y = f^*(X) + \varepsilon, \qquad \varepsilon \sim (0, \sigma^2). \tag{2.2}$$

Here, $f^* \colon \mathcal{X} \to \mathbb{R}$ is the true regression function (the Bayes-optimal predictor under squared loss) and ε is noise with 0 mean and variance σ^2. Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$ drawn i.i.d. from this model, we train a predictor $\hat{f}(x) = \hat{f}(x; \mathcal{D})$.

Definition 2.3 (Expected Test Risk)**.** The expected test risk (expected prediction error) at a test point x is

$$R(x) = \mathbb{E}_{\mathcal{D},\,\varepsilon}\Big[(\hat{f}(x) - Y)^2\Big], \tag{2.3}$$

where the expectation is taken over both the randomness of the training set $\mathcal{D}$ and the test noise ε.

Theorem 2.4 (Bias–Variance–Noise Decomposition of Risk)**.** *Under the model* (2.2)*, for any fixed test point x,*

$$\mathbb{E}_{\mathcal{D},\,\varepsilon}\Big[(\hat{f}(x) - Y)^2\Big] = \underbrace{\big(\mathbb{E}_{\mathcal{D}}[\hat{f}(x)] - f^*(x)\big)^2}_{\text{Bias}^2} + \underbrace{\mathbb{E}_{\mathcal{D}}\Big[(\hat{f}(x) - \mathbb{E}_{\mathcal{D}}[\hat{f}(x)])^2\Big]}_{\text{Variance}} + \underbrace{\sigma^2}_{\textit{Irreducible noise}}. \tag{2.4}$$

Proof. Since $Y = f^*(x) + \varepsilon$ and ε is independent of $\mathcal{D}$ with mean 0, we have:

$$\begin{aligned}\mathbb{E}\Big[(\hat{f}(x) - Y)^2\Big] &= \mathbb{E}\Big[(\hat{f}(x) - f^*(x) - \varepsilon)^2\Big] \\ &= \mathbb{E}\Big[(\hat{f}(x) - f^*(x))^2\Big] - 2\,\mathbb{E}\Big[(\hat{f}(x) - f^*(x))\varepsilon\Big] + \mathbb{E}[\varepsilon^2].\end{aligned} \tag{2.5}$$

By independence of ε and $\hat{f}(x)$, the cross-term vanishes: $\mathbb{E}[(\hat{f}(x) - f^*(x))\varepsilon] = \mathbb{E}[\hat{f}(x) - f^*(x)] \cdot \mathbb{E}[\varepsilon] = 0$. The last term is σ^2. Applying Theorem 2.2 to the first term with $\hat{\theta} = \hat{f}(x)$ and $\theta = f^*(x)$ gives the result. □

Note that the irreducible noise σ^2 is zero when the data is generated without noise, i.e. $Y = f^*(X)$. In this deterministic setting, the risk decomposes into only the first two terms: the sum between the Bias^2 and Variance. In reality, noise is almost surely always going to be present, and σ^2 sets a lower bound on achievable test error for any arbitrary estimator.

3. The Bias–Variance Tradeoff

The standard bias–variance tradeoff is often illustrated by a "U-shaped" curve of test error as a function of model complexity [2, 4].

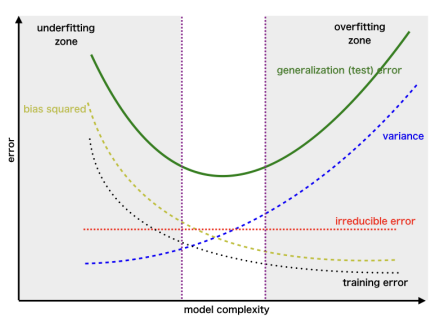

Figure 1. Example diagram illustrating typical bias, variance, and test error trends [7].

For very simple models, test error is dominated by high bias, since the hypothesis class is too rigid and limited, failing to capture the true function and thus causing underfitting. As model complexity increases, the bias decreases because the model becomes more capable of expressing the true data distribution. However, beyond a certain point, the variance begins to dominate as the model becomes overly sensitive to fluctuations, noise, and outliers in the training data. When our model is too sensitive, we see overfitting and increased test error. This produces the classical U-shaped curve, where the optimal model complexity balances bias and variance.

3.1. Connection to the Approximation–Estimation Error Tradeoff

This behaviour closely mirrors the approximation–estimation error tradeoff in statistical learning theory [2, 5]. The *approximation error* measures how well the hypothesis class $\mathcal{H}$ can represent the true function f^*; formally, it is determined by the distance between f^* and the best-in-class function $f^*_{\mathcal{H}} = \arg\min_{f\in\mathcal{H}} \mathbb{E}[(f(X) - f^*(X))^2]$. The *estimation error* reflects the difficulty of learning a good function from finite data — the gap between the learned $\hat{f}$ and the best $f^*_{\mathcal{H}}$.

Under suitable conditions, these two perspectives align with the bias–variance decomposition. In particular, if the learned predictor $\hat{f}(x)$ is a nearly unbiased estimator of $f^*_{\mathcal{H}}(x)$, the best possible function within the hypothesis class, then the following approximate correspondences hold.

Proposition 3.1 (Approximate Equivalence Under Near-Unbiasedness)**.** *Suppose that* $\mathbb{E}_{\mathcal{D}}[\hat{f}(x)] \approx f^*_{\mathcal{H}}(x)$ *for all* x*. Then:*

(i) The squared bias is approximately equal to the approximation error:

$$\mathrm{Bias}^2(x) = \big(\mathbb{E}[\hat{f}(x)] - f^*(x)\big)^2 \approx \big(f^*_{\mathcal{H}}(x) - f^*(x)\big)^2. \tag{3.1}$$

(ii) The variance term reflects the estimation error:

$$\mathrm{Var}(x) = \mathbb{E}\Big[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2\Big] \approx \mathbb{E}\Big[(\hat{f}(x) - f^*_{\mathcal{H}}(x))^2\Big]. \tag{3.2}$$

Proof. For (i): under the assumption $\mathbb{E}[\hat{f}(x)] = f^*_{\mathcal{H}}(x)$, substitute directly into the bias definition to obtain $\mathrm{Bias}^2(x) = (f^*_{\mathcal{H}}(x) - f^*(x))^2$, which is precisely the pointwise squared approximation error.

For (ii): if $\mathbb{E}[\hat{f}(x)] = f^*_{\mathcal{H}}(x)$, then $\mathrm{Var}(x) = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2] = \mathbb{E}[(\hat{f}(x) - f^*_{\mathcal{H}}(x))^2]$, which measures the variability of the learning algorithm around the best-in-class function. This is exactly the estimation error arising from the use of finite samples. □

Therefore, in regimes where the learning algorithm reliably approximates the best-in-class function, the bias–variance decomposition provides an interpretation that is consistent with the approximation–estimation error framework. However, this equivalence is only approximate and depends on the assumption that the estimator is close to unbiased with respect to $f^*_{\mathcal{H}}(x)$. When this assumption fails — for instance, in regularised estimators that are intentionally biased — the two decompositions no longer align cleanly.

4. Computational Illustration of the Bias–Variance Tradeoff

We now look to simulate the bias-variance tradeoff computationally using polynomial regression on synthetic data, using Monte Carlo averaging to approximate expectations.

4.1. **Experimental Setup**

We generate data from a known function with addition to Gaussian noise:

$$f^*(x) = \sin(2\pi x), \qquad x \sim \text{Uniform}(-1.5, 1.5), \qquad y = f^*(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \tag{4.1}$$

We use $n = 250$ training points, a noise value of $\sigma = 0.35$, and $n_{\text{test}} = 250$ fixed test points. For each polynomial degree $d \in \{1, 2, \ldots, 25\}$, we fit ordinary least-squares polynomial regression and record predictions on the test set. This process is repeated over 200 independent random seeds to approximate expectations.

Below, we see the true function $f^*(x) = \sin(2\pi x)$, along with three independent noisy training sets generated by the process above.

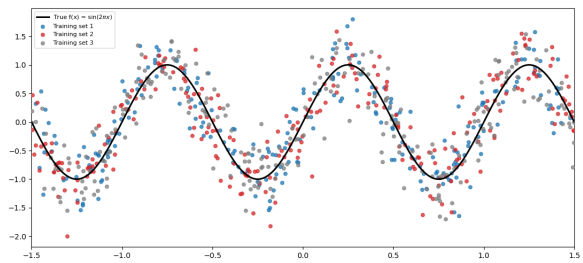

Figure 2. The true function $f^*(x) = \sin(2\pi x)$ and three training sets drawn from the data-generating process described above.

4.2. **Model Fits Across Complexity Levels**

Figure 3 overlays the predictions from all 200 seeds for three representative degrees. At $d = 5$ (left panel), the individual fitted curves cluster tightly together – variance is low – but they all miss the true function in the same systematic way, unable to capture the full oscillation of the sinusoid. This is bias in action: every training set produces a similar, and similarly wrong, prediction. At the near-optimal degree (centre panel), the mean prediction tracks f^* closely while the spread across seeds remains moderate, reflecting a balanced tradeoff. At $d = 25$ (right panel), the picture inverts. The mean prediction still approximates f^* reasonably well, but the individual curves scatter widely around f^* because each seed's polynomial contorts differently in attempt to fit its training set's noise. In this overly complex model regime, the model has enough flexibility to fit the true signal, but also enough to fit extra noise that happens to be present, producing predictions that are correct on average but unreliable for any single training set.

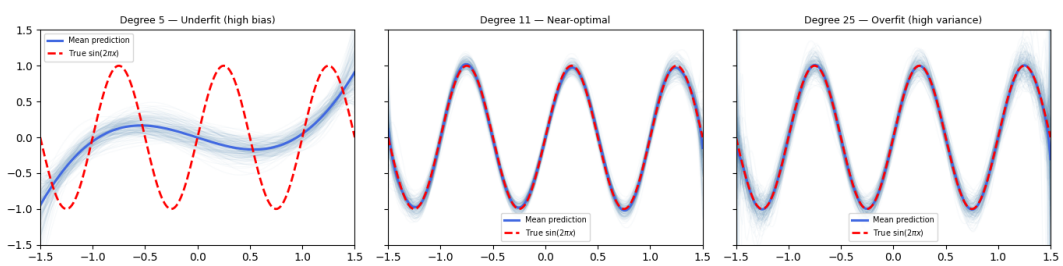

Figure 3. Fitted polynomials for three complexity levels. Light blue curves show individual seed fits; the dashed red curve is the true generating function $f^*(x) = \sin(2\pi x)$.

4.3. **Training and Test Error Curves**

Figure 4 plots training and test error as a function of polynomial degree. Training error decreases monotonically with complexity, as more expressive models have more parameters available to fit training data more accurately. This training error trend is a structural property: for nested hypothesis classes $\mathcal{H}_1 \subset \mathcal{H}_2 \subset \cdots$, the class $\mathcal{H}_{k+1}$ can always fit data better than or just as well as $\mathcal{H}_k$ (as long as the error was previously nonzero), so training error cannot increase with k. By contrast, test error exhibits the classic U-shape: it decreases as the model gains expressiveness, reaches a minimum at the optimal degree ($d = 11$ below), and then rises as the model begins to overfit.

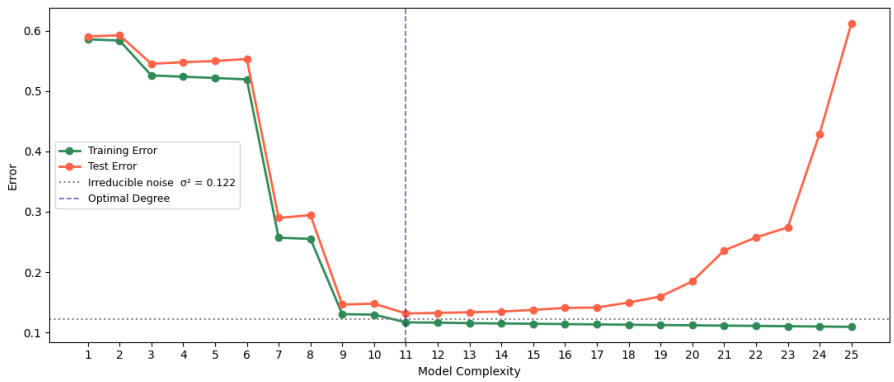

Figure 4. Training error (green) and test error (red) in relation to polynomial degree.

We analyze two crucial features of the training/test error plot. First, the training error falls below the noise level $\sigma^2 = 0.35^2 \approx 0.122$ at the optimal degree $d = 11$ and continues to drop as complexity increases further. In Ordinary Least Squares, if we write the fitted values as $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X^\top X)^{-1}X^\top$ the hat matrix and $\mathrm{tr}(H) = p$, then the expected residual sum of squares is

$$\mathbb{E}[\|\mathbf{y} - \hat{\mathbf{y}}\|^2] = \sigma^2(n - p), \tag{4.2}$$

where by n gives an expected training MSE of $\sigma^2(1 - p/n)$ [2]. A simple calculation using optimal $d = 11$ and a sample size of $n = 250$ gives an expected training MSE of roughly 0.117, exactly what we observe in Figure 4 as training error falls under $\sigma^2 \approx 0.122$ for the first time. Taking the limit $p \to n$ yields zero training error; At $p \approx n$, the model gains enough flexibility to interpolate every noisy point exactly.

Secondly, we see that test error stays above the noise level σ^2 throughout all polynomial degrees and approaches this floor near the optimal degree of $d = 11$. Note, our test MSE throughout the experiment is computed against noisy labels $y = f^*(x) + \varepsilon$ rather than the clean $f^*(x)$. The σ^2 term in the bias–variance decomposition (Theorem 2.4) represents noise that is inherently unpredictable from x – it reflects the information that was lost as y was corrupted. Thus σ^2 is a lower bound on test error for any predictor. A sufficiently flexible model can always reduce its training residuals past the test error bound σ^2 by fitting the noise in its own training set, which is precisely the overfitting process.

4.4. **Bias–Variance Decomposition**

Figure 5 shows the estimated squared bias, variance, and total test MSE in relation to model complexity. The result confirms the theoretical decomposition: bias decreases as the

model class increases in complexity, while the variance increases. Furthermore, The test MSE (red) lies perfectly on line bias^2 + variance + σ^2 (purple) by equivalence, and traces a U-shaped curve whose minimum identifies the optimal degree for a polynomial regression model.

Although the simulated graph qualitatively matches the classical bias-variance decomposition diagram, one minor difference is that bias^2 does not actually ever reach zero. This is because our true function $f^*(x) = \sin(2\pi x)$ cannot be entirely represented by any finite-degree polynomial. This is a case of mild model mis-specification, which we revisit in the context of double descent in Section 5.

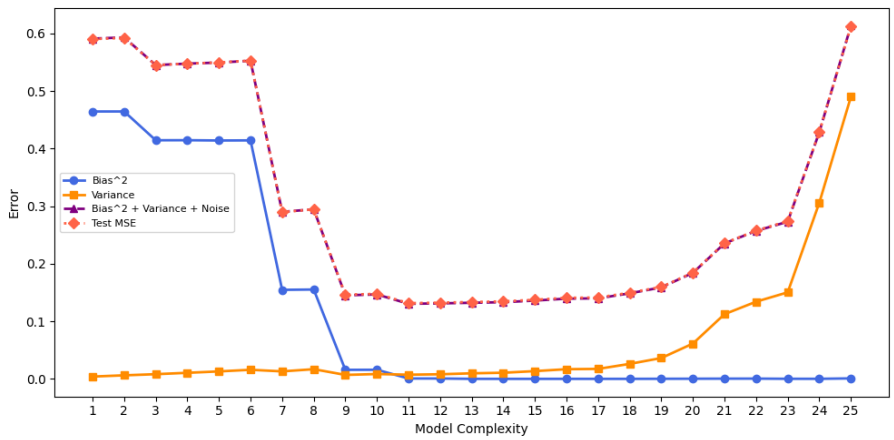

Figure 5. Resulting bias–variance decomposition.

4.5. **Sensitivity to Experimental Parameters**

Obtaining the desired U-shaped test error curve required appropriate choices of experimental parameters. In particular, the training sample size n and the noise level σ must be chosen so that bias and variance are sufficiently balanced across the range of degrees tested.

To illustrate the importance in the choice of training sample size, we ran an additional experiment with the same sinusoidal target function $f^*(x) = \sin(2\pi x)$ using 200 training sets of *only* $n = 15$ points. As shown in Figure 6, the test error does not exhibit the typical U-shape.

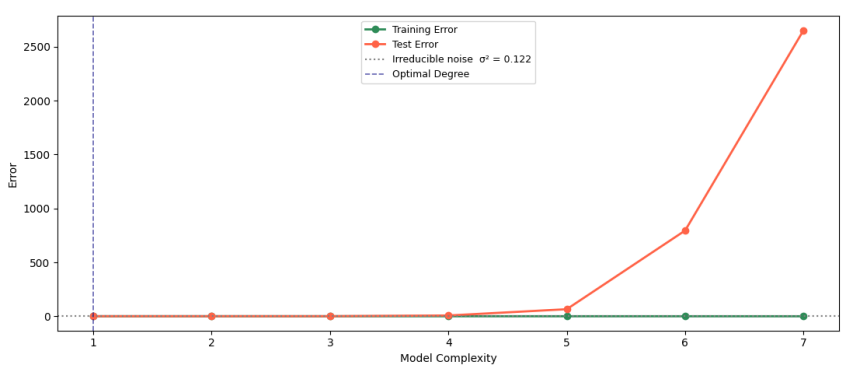

Figure 6. Poorly chosen dataset size parameter of $n = 15$. The conventional U-shaped test error curve does not appear; instead, variance dominates immediately and the optimal degree is trivially $d = 1$.

Instead, test error increases immediately with complexity, and the optimal fit is identified as $d = 1$ (a linear model). With so few training points relative to the number of polynomial coefficients, every additional degree adds more variance than it removes bias; the model does not have enough data to distinguish the true signal from the noise and thus, fitted curves vary drastically across seeds even at moderate complexity. The tradeoff we observe is effectively one-sided, where variance dominates at every level $d > 1$ and no genuine balance point exists within the range of degrees tested.

The noise parameter σ is equally important. Let us consider the structure of OLS polynomial regression. The fitted values take the form $\hat{\mathbf{y}} = H\mathbf{y}$, where $H = X(X^\top X)^{-1}X^\top$ is the hat matrix and X is the $n \times p$ design matrix of polynomial features [2]. Since $\mathbf{y} = \mathbf{f}^* + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$, the expected prediction is $\mathbb{E}[\hat{\mathbf{y}}] = H\mathbf{f}^*$, which depends only on the design matrix and the true function – not on σ. Therefore, the pointwise squared bias $(H\mathbf{f}^* - \mathbf{f}^*)_i^2$ is independent of the noise level. The variance of the i-th prediction is $\mathrm{Var}(\hat{y}_i) = \sigma^2 H_{ii}$ since $\hat{\mathbf{y}}$ is a linear function of the noise. Averaging over the n training points and using the identity $\mathrm{tr}(H) = p$ [2] gives the mean prediction variance as $\sigma^2 p/n$. The total expected risk at degree d is therefore

$$R(d) = \underbrace{\frac{1}{n}\|(H - I)\mathbf{f}^*\|^2}_{\mathrm{Bias}^2(d)\text{: independent of } \sigma} + \underbrace{\frac{\sigma^2 p}{n}}_{\mathrm{Variance}(d)\text{: linear in } \sigma^2} + \sigma^2. \tag{4.3}$$

Thus, increasing σ leaves the bias unchanged, but inflates both the variance and the irreducible noise floor, shifting the balance of the tradeoff. As σ becomes large, $\sigma^2 p/n$ dominates even at low polynomial degree (complexity), causing risk to increase monotonically with complexity. This eliminates the U-shaped risk curve, and gives the shape seen in Figure 6. Conversely, as $\sigma \to 0$, variance becomes negligible and the risk decreases monotonically as bias shrinks.

5. Double Descent

We have derived and observed that the classic bias-variance tradeoff predicts that test error follows a U-shaped curve as a function of model complexity: high test error for underfitted models with large bias that eventually reaches a minimum, then rises again as variance from overfitting begins to dominate. This expected behaviour was the dominant framework for thinking about generalization for decades, codified in Hastie, Tibshirani & Friedman [2] and illustrated by the common bias-variance diagrams.

An observation by Belkin et al. [1] challenged this picture: when model complexity is pushed beyond the *interpolation threshold* – the point at which the model has enough parameters to fit the training data exactly – test error does not continue to increase. Instead, it decreases again, eventually falling below the original minimum. This is the *double descent* phenomenon.

This section reproduces the double descent phenomenon computationally in two settings: through polynomial regression as seen in Section 4 and the Gaussian feature model. We also explore the effect and mechanism of both explicit and implicit regularization. In this section, expectations are approximated using Monte Carlo averaging over many random seeds.

5.1. Experiment 1 – Double Descent in Polynomial Regression

We generate synthetic data using a cosine true function with added noise:

$$f^*(x) = \sin(3\pi x), \quad x \sim \mathrm{Uniform}(-1, 1), \quad y = f^*(x) + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 0.5, \tag{5.1}$$

with $n = 20$ training samples and $n_{\text{test}} = 500$ held-out test points. Polynomial regression of degree $d \in \{1, \ldots, 35\}$ serves as the model class, with the interpolation threshold observed below at $d = 19$. For $d > 19$ the model is over-parametrised and in this regime, we fit the minimum-norm interpolating solution via ridge regression with $\alpha \to 0$ ($\alpha = 10^{-10}$). All statistics are averaged over 150–200 independent random seeds.

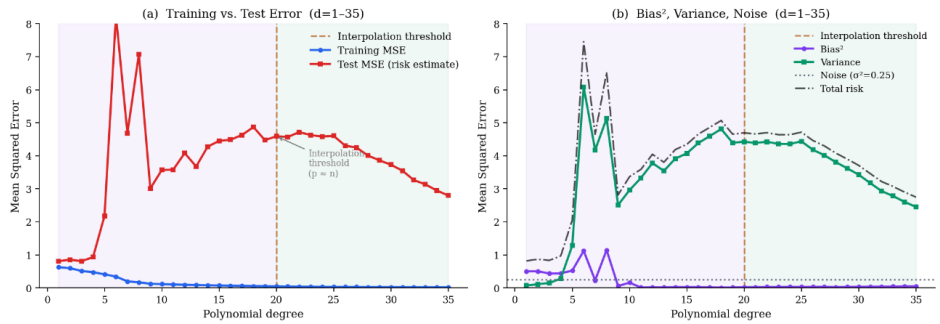

Figure 7. Left: Training error (blue) and test error (red) in relation to model complexity. Right: Bias^2 (purple), Variance (green), irreducible noise $\sigma^2 = 0.25$ (grey dotted), and test error (total risk) Bias^2+Var+σ^2 (black dash-dot) in relation to model complexity.

In the above figure, we observe double descent. Test error in the under-parametrized regime ($d < 19$) is clearly U-shaped. This U-shape required careful tuning of the experimental setup: with $n = 20$ training samples and $\sigma = 0.5$ noise, the bias and variance terms are sufficiently balanced that neither dominates across the full range of degrees. The U-shape collapses once we enter the over-parametrized regime (past a polynomial degree of $d = 19$), as test error begins to descend once again..

In summary, while in the under-parametrized regime ($d < 19$), the resulting graphs match the classic sketches for the bias-variance decomposition and error curves. At the interpolation threshold ($d = 19$), training error nearly reaches zero, while test error spikes sharply. Entering the over-parametrized regime ($d > 19$), test error begins to decrease. Variance falls because the minimum-norm solution in a higher-dimensional space is "smoother" and less sensitive to individual training points – this is a manifestation of implicit regularization. The net effect is a clear double descent in test error.

In the figure below, we widen the scope of this trend by extending the model complexity sweep past $d = 35$ and observe a further decrease in test error as fitted polynomial models reach a degree of $d = 80$.

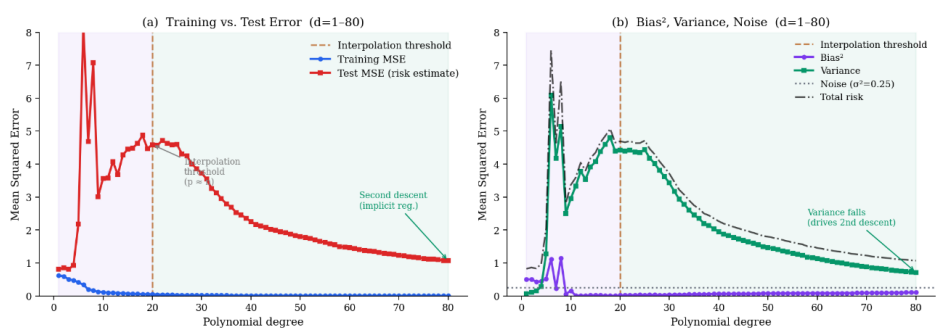

Figure 8. Error and bias-variance curves with the same parameters used in Figure 7, but sweeping model complexity past $d = 35$.

5.2. **Experiment 2 – Double Descent with the Gaussian Feature Model**

To reproduce the theoretical setting of Belkin et al. [1] and Hastie et al. [3], we implement the *Gaussian feature model*, allowing us to vary the parameter-to-sample ratio p/n continuously. We generate data using the following setup:

$$y = Xw^* + \varepsilon, \quad X \in \mathbb{R}^{n\times p},\ X_{ij} \sim \mathcal{N}\Big(0, \tfrac{1}{p}\Big), \quad \varepsilon \sim \mathcal{N}(0, \sigma^2), \quad \sigma = 1, \tag{5.2}$$

where $w^* \in \mathbb{R}^p$ is a sparse signal ($w_i^* = 1/5$ for $i \le 5$, zero otherwise). We fix training size at $n = 50$, let p range from 2 to 200, giving $p/n \in [0.04, 4.0]$. For each p, we draw $X_{\text{train}} \in \mathbb{R}^{50\times p}$ and $X_{\text{test}} \in \mathbb{R}^{500\times p}$, then fit ridge regression with regularization α. The ridgeless case uses $\alpha = 10^{-10}$. Results are averaged over 80 independent seeds.

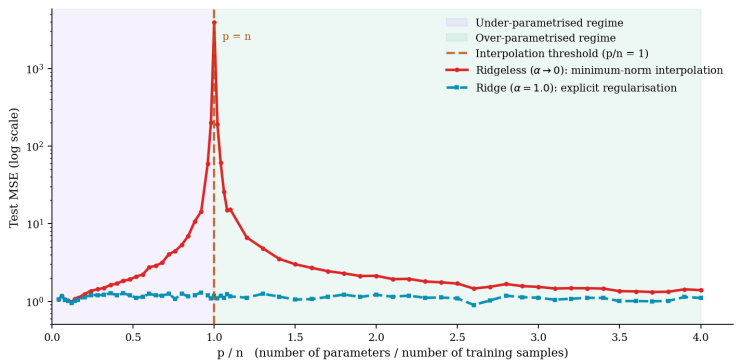

Figure 9. Test MSE in relation to p/n ratio in the Gaussian feature model.

The ridgeless curve reproduces the shape reported by Belkin et al. [1]: a U-shaped test error in the under-parametrized regime, divergence at $p = n$, and an eventual decrease for $p > n$. We observe that past $p/n \ge 3$, the ridgeless estimator's test error falls *below* the original minimum, as we have many more parameters available for increased generalization.

5.3. **Experiment 3 — The Mitigating Role of Regularisation**

We observe the effect of explicit ridge regularization on the double descent peak. Although we see the contrast between test error curves of drastically different ridge levels in experiment 2, Figure 10 shows test error curves for ridgeless regression and three increasing values of $\alpha \in \{0, 0.1, 1.0, 10.0\}$.

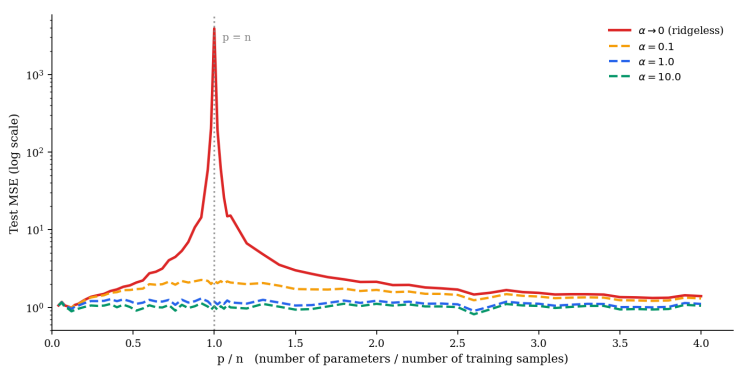

Figure 10. Effect of ridge regularization α on the double descent curve using log-scaled y-axis to display full peak.

As α increases, the divergence peak lowers. For instance, we observe that for $\alpha = 0$, test error experiences its typical peak, while for $\alpha = 10$, no peak is visible and the curve is essentially monotone. This confirms a key theoretical result from Hastie et al. [3]: any positive regularisation smooths out the singularity at $p/n = 1$.

5.4. **Double Descent and Implicit Regularization**

We explore why the second descent in test error occurs in the context of implicit regularization. When $p > n$, the system $Xw = y$ is under-determined: we have infinitely many weight vectors that interpolate the training data. Minimum-norm least squares selects the unique solution with smallest $\|w\|_2$, given by the Moore–Penrose pseudoinverse:

$$\hat{w} = X^{\dagger} y = X^{\top}(XX^{\top})^{-1} y. \tag{5.3}$$

This selection is a form of implicit regularization: without any explicit penalty, the optimization process chooses solutions with a smaller norm. This is similar to what ridge regression achieves with an explicit ℓ_2 penalty, but here it arises purely from the choice of solution in the under-determined system.

Geometrically, when $p \gg n$, the feature vectors X_i are nearly orthogonal in a high dimension. Since the p-dimensional space is much larger than the number of constraints, the least-norm solution can fit the training labels using small coefficient values spread across many dimensions. This allows predictions on new test points to remain close to the true signal, since small coefficients help prevent a single noisy training point or outlier from dominating the prediction. As p increases, the solution better distributes its weight across many small coefficients. This reduces sensitivity to individual training noise and consequently reduces overall variance.

This explains the double descent: at the interpolation threshold ($p = n$), there is exactly one solution that fits our data, and it is forced to use large coefficients to simultaneously satisfy all n constraints, causing variance to spike. When $p/n > 1$, the solution space expands and the minimum-norm criterion can select smoother solutions, causing variance to fall while bias remains unchanged – this contributes to the second descent of the test error curve.

5.5. **Summary of Experimental Findings**

Table 1 summarizes the behaviour of key quantities across the three regimes.

Aspect	**Under-param. ($p < n$)**	**Interpolation ($p = n$)**	**Over-param. ($p > n$)**
Training error	Decreases with complexity	Collapses to zero	Stays at zero
Test error (risk)	U-shaped; classic B-V tradeoff	Peak / divergence	Decreases: double descent
Bias^2	Decreases as model improves	Low but non-zero	Roughly stable
Variance	Increases with complexity	Spikes sharply	Decreases (implicit reg.)
Implicit reg.	–	None; pure interpolation	Minimum-norm selection
Effect of ridge α	Modest improvement	Eliminates peak	Raises floor; smoother curve

Table 1. Summary of experimental findings across the three regimes of model complexity.

5.6. **Further Research on Double Descent**

The referenced paper from Belkin et al. [1] documented the double descent phenomenon empirically and proposed that minimum-norm interpolation is the key mechanism underlying the second descent. In Hastie et al. [3], the phenomenon was further explored theoretically, where random matrix theory was used to derive exact closed-form risk expressions for

ridgeless and ridge regression. Their analysis confirmed that the test error curve has the double descent shape for ridgeless regression at any fixed $\gamma \neq 1$, with a divergence precisely at $\gamma = 1$, and that any positive ridge penalty $\lambda > 0$ smooths out the singularity – consistent with what was observed in Experiment 3.

6. Conclusion

This paper has derived the foundational bias-variance decomposition, illustrated classical error and bias-variance behaviour, and explored the double descent phenomenon. We derived the classical three-term risk decomposition (Theorem 2.4), drew its connection to the approximation-estimation error tradeoff (Proposition 3.1), and validated both frameworks computationally through polynomial regression experiments.

While extending this analysis into over-parameterized regimes, we reproduced the double descent phenomenon in two complementary settings: polynomial regression and the Gaussian feature model. Our experiments confirm the key findings of Belkin et al. [1] and the theoretical predictions of Hastie et al. [3], as we observed the risk curve spike at the interpolation threshold and its second decrease, driven by the implicit regularization of minimum-norm interpolation. Additionally, we demonstrated that explicit ridge regularization ultimately mitigates the test error peak at the interpolation threshold.

Further exploration of these topics may include experimenting with non-polynomial predictors. In particular, the double descent phenomenon extends beyond models of this type; Nakkiran et al. [6] demonstrated that deep neural networks exhibit similar double descent behaviour as a function of model width, training epochs, and dataset size. This suggests that the phenomenon is a general property of over-parametrized learning rather than an aspect of specific model classes.

The results and analyses in this paper illustrate that the classical bias-variance tradeoff, while providing a foundational guide for model selection, is incomplete in the regime of over-parametrized models.

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